

CO 250 Winter 2026: Midterm practice problems

These are test problems from previous terms, no solutions will be provided. It is not an indication of the length or the difficulty of your actual test. You are welcome to discuss solutions to these problems on Piazza and during office hours.

M-1. A small textile company sells 3 types of fabrics. They purchase pre-processed fabric and then dye and finish the fabric at the facility. Each fabric takes a certain amount of time to be dyed and finished and has a different purchase and selling price, which are illustrated in the table below. Here, the purchase price refers to the price at which the company buys the pre-processed fabric, and the selling price refers to the price at which the company sells the fabric.

Fabric	Purchase Price (\$/m)	Selling price (\$/m)	Dyeing (hrs/m)	Finishing (hrs/m)
A	3	7	4	3
B	1	4	1	2
C	1	5	3	1

The dyeing and finishing facilities can only be used for at most 700 hours and 800 hours respectively in the given time frame.

- (a) Formulate an LP that maximizes the profit generated by the company.

Provide a brief description of your variables, objective function and constraints.

- (b) Suppose each of the facilities has an operating and running cost. The dyeing facility costs 1 dollar for every hour it is being used, and the finishing facility costs .7 dollars for every hour. Modify the LP in the previous part to reflect these costs.

Provide a brief description of any new variables, changes in the objective function and any new constraints. You do not need to copy the parts of your LP that stay unchanged.

- (c) Suppose the operating and running costs from part M-1.b are charged by the hour, i.e. the textile company will be charged the full per hour cost even if the company uses the facilities for a part of it. Modify your LP into an IP to incorporate this condition.

Provide a brief description of any new variables, changes in the objective function and any new constraints. You do not need to copy the parts of your LP that stay unchanged.

- (d) In addition to the provided constraints, suppose the company can only sell fabric B if it is selling fabric C. Modify your IP to incorporate this constraint.

Provide a brief description of any new variables, changes in the objective function and any new constraints. You do not need to copy the parts of your LP that stay unchanged.

M-2. The company CornCO has three factories, each making a different type of corn products: corn oil (O), corn syrup (S), and corn flakes (F). It takes $\alpha_O, \alpha_S, \alpha_F$ kilograms of corn to produce one kilogram of O, S, F, respectively. Each kilogram of O, S, F will sell for $\$p_O, \$p_S, \$p_F$ respectively.

The company sources its corn from three different towns: Arthur (A), Berwick (B) and Clifford (C). The towns A, B, C produce $\beta_A, \beta_B, \beta_C$ kilograms of corn, respectively. For each product $i \in \{O, S, F\}$ and each town $j \in \{A, B, C\}$, it costs $\$q_{ij}$ to buy and transport one kilogram of corn from town j to factory i .

(You may assume that $\alpha_i, p_i, \beta_j, q_{ij}$ are given constants. We allow fractional kilograms to be used.)

- (a) Formulate this problem as a linear program, with the goal of maximizing total profit. Clearly define your variables, clearly identify your objective function and constraints, and briefly explain your constraints.
- (b) Suppose the company has a contract to produce at least M kilograms of corn syrup (S). But if less than M kilograms are produced, then the company will need to pay a penalty of $\$r$ per kilogram less than M produced. Explain how you would change the LP from part (a) to implement this new condition. Explain why your method works.

M-3. Suppose we are formulating a certain integer program problem, which includes these five variables x_1, x_2, x_3, x_4, x_5 . We already have the following constraints in the IP:

$$x_1, x_2, x_3 \in \{0, 1\}, \quad x_4 \geq 0, \quad x_4 \leq 100, \quad x_5 \geq 0, \quad x_4, x_5 \in \mathbb{Z}$$

Each of the following is a condition that we want to implement for our IP. Write down constraints that formulate each condition. You may create additional integer variables. Explain why your formulation works. (Note that these conditions are independent of each other.)

- (a) At least two of x_1, x_2, x_3 have value 1.
- (b) If $x_1 = 1$, then $x_4 \leq 30$.
- (c) x_5 is divisible by 3.
- (d) Exactly one of the following holds:
 - The sum of x_4 and x_5 is even.
 - At least one of x_2 and x_3 is equal to 0.

M-4. Suppose we are formulating a non-linear program, and three of the variables are x_1, x_2, x_3 , which can take real values. For each of the following conditions, write down constraint(s) that formulate the condition, and give a brief explanation. You may only use polynomial and/or trigonometric functions in your constraints.

(a) At least two of the three variables x_1, x_2, x_3 are equal.

Hint: It is possible to use only one constraint to do this.

(b) All three variables x_1, x_2, x_3 are different (i.e. $x_1 \neq x_2, x_1 \neq x_3, x_2 \neq x_3$).

M-5. Consider the following Linear Program.

$$\begin{aligned} \min \quad & (-1, 2, 3)x \\ \text{s.t.} \quad & \begin{pmatrix} 1 & -1 & 7 \\ 4 & 3 & -3 \\ 1 & -2 & 9 \end{pmatrix} x \begin{matrix} \leq \\ \geq \\ = \end{matrix} \begin{pmatrix} 15 \\ -30 \\ 17 \end{pmatrix} \\ & x_1 \leq 0, \ x_2 \text{ free}, \ x_3 \geq 0 \end{aligned} \quad (P)$$

(a) Convert the LP (P) into Standard Equality Form (SEF).

(b) You are given a feasible solution $\bar{x} = (-3, -1, 2)^T$ to the LP (P). Derive a feasible solution to the LP in SEF, that corresponds to $\bar{x}$. Find the value of the derived feasible solution.

M-6. (a) Determine whether $\bar{x} = (1, 0, 0, 0)^T$ is a basic feasible solution for the following Linear Program. If it is a basic feasible solution provide a corresponding basis.

$$\begin{aligned} & \begin{pmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 0 & 5 \\ 0 & 1 & 1 & 6 \end{pmatrix} x = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \\ & x \geq \mathbf{0} \end{aligned}$$

(b) Determine all values of the parameters α, β such that $\bar{x} = (3, 0, 2, 0, 0)^T$ is not a basic solution corresponding to $\{1, 3, 4\}$, but is a basic solution corresponding to the basis $\{1, 3, 5\}$ for the following Linear Program. Provide a brief justification.

$$\begin{aligned} & \begin{pmatrix} 1 & 2 & 0 & 3 & \beta \\ 2 & 4 & 0 & \alpha & 4 \\ 0 & 1 & 1 & 6 & 1 \end{pmatrix} x = \begin{pmatrix} 3 \\ 6 \\ 2 \end{pmatrix} \\ & x \geq \mathbf{0} \end{aligned}$$

- M-7. (a) Determine whether or not the given LP is unbounded, infeasible, or has an optimal solution. If it is unbounded, also give a feasible solution whose objective value is at least 1000. Give a self-contained proof of your answer (i.e. your proof should not depend on theorems from class or the textbook). Some vectors are given for the LP that may or may not help as possible certificates.

$$\begin{aligned} \max \quad & (4, 1, 0, 0, 1, -9, -10)x \\ \text{s.t.} \quad & \begin{pmatrix} 2 & -2 & 0 & 1 & -1 & -3 & -5 \\ 7 & -3 & 2 & 2 & -2 & 1 & -8 \end{pmatrix} x = \begin{pmatrix} 5 \\ 9 \end{pmatrix} \\ & x \geq \mathbf{0} \end{aligned}$$

Vectors to consider: $\bar{x} = (1, 1, 0, 0, 0, 5, 9)^T$, $\bar{y} = (-2, 1)^T$, $d = (0, 0, 0, 1, 1, 0, 0)^T$, $\tilde{x} = (1, 0, 2, 0, 0, 3, -1)^T$, $\tilde{y} = (-3, 1)^T$.

- (b) Alice is interested in an outcome of an LP in SEF, but only some of the entries of the LP are revealed to Alice. Entries marked with $\star$ are revealed while entries marked with $?$ are not known to Alice. Alice never makes mistakes. Alice tells us that she already knows the outcome of the LP. We do not know the values of the entries revealed to Alice.

$$\begin{aligned} \max \quad & (\star, \star, \star, \star, \star)x \\ \text{s.t.} \quad & \begin{pmatrix} ? & ? & ? & ? & ? \\ ? & ? & ? & ? & ? \end{pmatrix} x = \begin{pmatrix} \star \\ \star \end{pmatrix} \\ & x \geq \mathbf{0} \end{aligned}$$

Only one option is correct.

- i. From the information, we can conclude that the LP is unbounded.
- ii. From the information, we can conclude that the LP is infeasible.
- iii. From the information, we can conclude that the LP has an optimal solution.
- iv. From the information, we cannot determine the outcome of the LP.
- v. None of the above options is correct.

- M-8. Let (P) be the linear program $\max\{c^T x + \bar{z} : Ax = b, x \leq 0\}$.

The following are choices for three possible conditions.

- (α) $d \geq 0, d \leq 0$.
- (β) $Ad \geq 0, Ad = 0, Ad \leq 0$.
- (δ) $c^T d > 0, c^T d = 0, c^T d < 0$.

Select the correct conditions (α), (β), (δ) so that the following statement is true: If there exist a feasible solution $\bar{x}$ and a vector d such that (α), (β), (δ), then (P) is unbounded.

Prove this statement. You may not use results from class.

- M-9. (a) Bring the following LP into canonical form for the basis $B = \{1, 2\}$. Find the corresponding basic solution and its objective function value.

$$\begin{aligned} \max \quad & (3, 5, 2, 7, 4)x \\ \text{s.t.} \quad & \begin{pmatrix} 2 & 1 & 3 & 1 & 2 \\ 1 & 1 & 2 & 3 & 1 \end{pmatrix} x = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \\ & x \geq \mathbf{0} \end{aligned}$$

- (b) Carol says that there is a way to fill numbers in the boxes of the below LPs such that both become canonical forms of the same LP. Bob says that it is impossible to do so. Determine who is right. Justify your answer.

$$\begin{aligned} \max \quad & \left(5, -1, \boxed{}, \boxed{}, \boxed{}, \boxed{} \right) x \\ \text{s.t.} \quad & \begin{pmatrix} 4 & -1 & \boxed{} & \boxed{} & \boxed{} & \boxed{} \\ -4 & 1 & \boxed{} & \boxed{} & \boxed{} & \boxed{} \end{pmatrix} x = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \\ & x \geq \mathbf{0} \end{aligned}$$

and

$$\begin{aligned} \max \quad & \left(0, 0, \boxed{}, \boxed{}, \boxed{}, \boxed{} \right) x + \boxed{} \\ \text{s.t.} \quad & \begin{pmatrix} \boxed{} & \boxed{} & \boxed{} & \boxed{} & \boxed{} & \boxed{} \\ \boxed{} & \boxed{} & \boxed{} & \boxed{} & \boxed{} & \boxed{} \end{pmatrix} x = \begin{pmatrix} -1 \\ 2 \end{pmatrix} \\ & x \geq \mathbf{0} \end{aligned}$$

- M-10. We are using the simplex method to solve a linear program, and we currently have the following LP in canonical form.

$$\begin{aligned} \max \quad & (0 \quad -3 \quad 0 \quad 2 \quad 0 \quad 0 \quad 5) x + 10 \\ \text{s.t.} \quad & \begin{pmatrix} 1 & 3 & 0 & 2 & 0 & 0 & -8 \\ 0 & -6 & 1 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 1 & 0 & 2 \\ 0 & 5 & 0 & -5 & 0 & 1 & 3 \end{pmatrix} x = \begin{pmatrix} 8 \\ 9 \\ 12 \\ 0 \end{pmatrix} \\ & x \geq \mathbf{0} \end{aligned}$$

- Write down the current basis and basic feasible solution. (No justifications required.)
- For the current iteration of simplex, determine all possible entering variables, and for each of them, determine all possible leaving variables. (We are not using Bland's rule here. Any eligible entering/leaving variables should be listed.)
- If we use Bland's rule for selecting the entering and leaving variables, which ones would we choose?
- For the choices you have made in part (c), determine the new basic feasible solution and its objective value. (Do not calculate the canonical form.)

M-11. Run the Simplex algorithm on the following LP, starting with the basis $B = \{1, 4\}$. Use Bland's Rule when you have multiple choices of entering or leaving variables. You may use the method of canonical forms, dictionaries, or the tableau method.

In an iteration, you should show how you are determining the entering and leaving variables, and state the new basis and basic feasible solution. You may then directly record the canonical form, dictionary, or tableau associated with the new basis without showing work.

For this problem, keep non-integer values as exact fractions.

If the problem is unbounded, give the certificate of unboundedness.

Stop after at most three iterations of Simplex Method.

$$\begin{aligned} \max \quad & (0 \ 1 \ 1 \ 0) x \\ \text{s.t.} \quad & \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{pmatrix} x = \begin{pmatrix} 3 \\ 3 \end{pmatrix} \\ & x \geq \mathbf{0} \end{aligned}$$

M-12. Let (P) be the following linear program.

$$\begin{aligned} \max \quad & (1 \ 2 \ 3) x \\ \text{s.t.} \quad & \begin{pmatrix} -1 & 1 & 0 \\ -3 & 1 & -1 \end{pmatrix} x = \begin{pmatrix} 2 \\ 5 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

Suppose we run Phase I of the two-phase simplex, which results in the following.

$$\begin{aligned} \max \quad & (-2 \ 0 \ -1 \ -2 \ 0) x - 3 \\ \text{s.t.} \quad & \begin{pmatrix} -1 & 1 & 0 & 1 & 0 \\ -2 & 0 & -1 & -1 & 1 \end{pmatrix} x = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

What can you conclude about (P)? If it is feasible, then provide a basic feasible solution for (P) and write (P) in canonical form for your basis. If it is infeasible, then provide a certificate of infeasibility, and prove that your certificate is correct.

M-13. Let (P) be the linear program $\max\{c^T x : Ax = b, x \geq 0\}$ where A is an $m \times n$ matrix with full row rank, $b \geq 0$, and $x = (x_1, \dots, x_n)^T$. Let (AuxP) be the auxiliary linear program for (P) with auxiliary variables $x' = (x_{n+1}, \dots, x_{n+m})^T$. Suppose (AuxP) has optimal value 0.

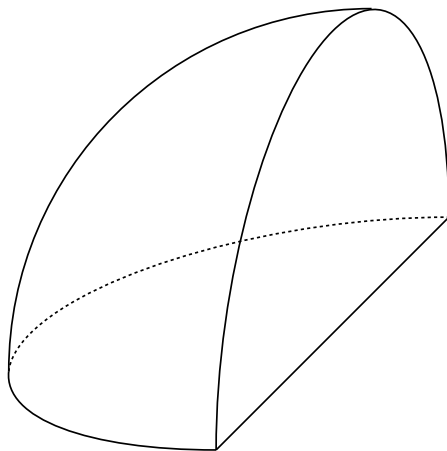
Prove that if (AuxP) has a basic optimal solution with basis B where $B \cap \{n+1, \dots, n+m\} \neq \emptyset$, then (P) has a basic feasible solution $\bar{x}$ with basis $B' \subseteq \{1, \dots, n\}$ where $\bar{x}_i = 0$ for some $i \in B'$.

M-14. Consider the set

$$S = \{x \in \mathbb{R}^2 : x_1 - x_2 \leq 3 \text{ OR } x_1 + x_2 \geq 5\}.$$

Is S a convex set? Give a proof for your answer.

- M-15. (a) The following picture depicts a quarter part of the unit ball in $\mathbb{R}^3$. In the picture, indicate clearly all extreme points.



- (b) Consider the polyhedron

$$P = \left\{ x \in \mathbb{R}^3 : \begin{pmatrix} 3 & 1 & 4 \\ 1 & 5 & 9 \\ 2 & 6 & 5 \\ 3 & 5 & 8 \\ 9 & 7 & 9 \end{pmatrix} x \leq \begin{pmatrix} 8 \\ 20 \\ 13 \\ 17 \\ 26 \end{pmatrix} \right\}$$

and the point $\bar{x} = (1, 1, 1)^T$. Is $\bar{x}$ an extreme point of P ? Justify your answer.